

Building Azure Infrastructure

Put it all together — provision real resource groups and networking with the azurerm provider.

azurerm patterns

Reading the docs

Naming & tags

Networking

Lab 06

Today's roadmap



1 • The target architecture

What we'll build over the next few sessions.



2 • Reading provider docs

Turn a registry page into working HCL, fast.



3 • Naming & tags

Conventions that keep a shared subscription sane.



4 • Lab 06

Create resource groups & a virtual network.

01

The Target Architecture

One app, built across sessions 06–08.

What we'll provision



Resource groups

Separate groups for network, app and database.



Virtual network

A VNet with subnets for the workloads.



App Services

Frontend & backend on an App Service plan.



PostgreSQL

A managed database for the backend.



Key Vault

Secrets — imported in Session 08.



Tags everywhere

environment · owner · type on every resource.

Resource groups → network

RG
net-rg
networking



VNet
vnet + subnets
address space



RG
app-rg · db-rg
ready for S07



The lab gives you the **properties** — you write the HCL by reading the docs. That's the real skill.

02

Working with azurearm

Docs, conventions, and a resource-group example.

Reading provider documentation

Every resource has a registry page with **Arguments**, **Attributes** and examples.

- Required vs. optional arguments
- Attributes you can reference later (e.g. `.id`)
- Copy the example, then trim to what you need

HCL

```
resource "azurerm_resource_group" "net" {  
  name      = "${local.prefix}-net-rg"  
  location = "westeurope"  
  
  tags = {  
    environment = "prod"  
    owner       = var.owner  
    type        = "network"  
  }  
}
```

A virtual network & subnet

HCL

```
resource "azurerm_virtual_network" "main" {
  name          = "${local.prefix}-vnet"
  location      = azurerm_resource_group.net.location
  resource_group_name = azurerm_resource_group.net.name
  address_space = ["10.0.0.0/16"]
}

resource "azurerm_subnet" "app" {
  name                = "app"
  resource_group_name = azurerm_resource_group.net.name
  virtual_network_name = azurerm_virtual_network.main.name
  address_prefixes    = ["10.0.1.0/24"]
}
```

- The subnet **references** the VNet — Terraform infers the create order
 - Reuse `azurerm_resource_group.net` for name & location — never hard-code twice
 - `address_space` is the VNet's CIDR; each subnet takes a slice of it
-  Add subnets with `for_each` (Session 04) once the pattern is clear.

Naming & tags conventions



Prefix everything

Use your assigned `<prefix>` so names never collide in the shared subscription.



Consistent tags

environment · owner · type — set them from a `local.common_tags` map.



Region for the whole course: `westeurope`. Keep it consistent to simplify quotas and networking.

Applying Azure infrastructure

BASH

```
terraform fmt
terraform init
terraform validate
terraform plan -out infraPlan
terraform apply infraPlan
```

💰 Real Azure resources cost money. Always read the plan, and **destroy** what you don't need between sessions.

 LIVE DEMO

Provision infra in Azure

Build resource groups, a virtual network, App Services and a PostgreSQL database — the full

`fmt → init → validate → plan → apply` loop.

Demo · 06 **Provider** · azurerm

Resource groups & virtual network

Given only the properties, author the HCL for three tagged resource groups and a VNet — then apply and push to your course repo.

You'll practise: reading docs · tags via locals · `<prefix>` naming · plan files

Key takeaways

✓ Docs are the skill

Turn a registry page into working HCL.

✓ Conventions scale

Prefixes & shared tags keep things tidy.

✓ Mind the cost

Read the plan; destroy what you don't need.

✓ Next up

Session 07 — advanced resources.



Questions?

Up next — [Session 07: Advanced Resources.](#)